
EMCH 367 / ELCT 331

Modern Control Systems for Physical Engineering Applications

University of South Carolina

Fall 2026

Cross-listed: EMCH 367 / ELCT 331

Meeting Time: Tue/Thu 10:05–11:20 AM

Location: TBD

Instructor: J.C. Vaught

Office: TBD

Office Hours: TBD

Catalog Description

This course develops practical competency in modeling, feedback control, estimation, and implementation for real engineered systems. The course is concept-first in lecture and application-heavy in assignments. Students learn to build control-ready models of physical systems, design and evaluate feedback controllers, implement controllers digitally, handle key nonlinear effects, and work with multivariable systems using state space methods. The course includes two team projects with physical demonstrations and presentations.

Prerequisites

Students are expected to be fluent in calculus through multivariable calculus, differential equations, and linear algebra. Students must be comfortable writing and debugging code in Python or MATLAB and producing plots and numerical summaries. The course does not include a formal review of math prerequisites.

Required Resources

Students need a laptop capable of running the course programming environment. The course will standardize on a primary environment (Python or MATLAB) and provide a reference environment for grading. Students must be able to run provided starter code, produce required plots, and submit reproducible results. Hardware resources and safety requirements for demos are provided by the course and described on the course site.

Instructional Approach

Lecture emphasizes concepts, design reasoning, and engineering workflows rather than step-by-step derivations. Mathematical derivations and deeper proofs are provided through optional typed notes and video resources. Theory homework requires formal engagement with selected mathematical tools, while application homework emphasizes implementation, validation, and interpretation.

Assessment Weights

This course is not curved. Final grades are determined by the following components.

Component	Weight
Homework (11 sets, lowest dropped)	50%
Projects	40%
Project 1	15%
Project 2	25%
Exams (two in-term, 5% each)	10%

Grading Scale

Final letter grades are assigned according to the following scale.

Grade	A	B+	B	C+	C	D+	D	F
Percentage	90–100	85–89	80–84	75–79	70–74	65–69	60–64	<60

Homework

There are 11 homework sets total. The lowest score is dropped. The remaining 10 are weighted by rank, with the lowest grade receiving the highest weight.

Rank	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
Weight	10%	8%	6%	6%	4%	4%	3%	3%	3%	3%

HW0 is a mathematics prerequisites assessment due Sunday at midnight before the first day of class. This diagnostic helps students identify gaps in their mathematical preparation. HW1–HW10 each consist of two files: HW_X_m (mathematics/theory portion) and HW_X_c (computational/application portion).

Submission Requirements

HW0 is due Sunday at midnight before the first class meeting. HW1–HW10 are due Monday at midnight before that week's classes. Typed submissions only—handwritten work is not accepted; at minimum, all final answers must be typeset (e.g., L^AT_EX, Word with equation editor, or equivalent). Coding portions are graded on final output matching spec exactly, and all code submissions must be reproducible.

Exams

Two in-term exams (5% each) focus on applied reasoning: selecting modeling assumptions, choosing control structure, interpreting behavior, specifying engineering steps, and diagnosing failures. Each exam includes an optional challenge section worth up to 10% bonus on that exam.

The final exam is optional and covers material from all aspects of the course. Students who take the final exam may choose to apply their score as a replacement for either one homework grade or one in-term exam grade—whichever benefits the student most. Alternatively, students may choose not to apply the final exam score at all. The final exam cannot hurt your grade; it can only help.

Additional Credit

Students may earn additional credit of up to 5% per assignment or project by demonstrating work that exceeds expectations. An optional research project may also be approved until two weeks before the final day of class. A 5-point research project is expected to involve a focused investigation of a specific control technique or application, with a brief written summary and demonstration. A 10-point research project is expected to involve a more comprehensive study, including implementation, experimentation, and a detailed written report with analysis of results.

Learning Outcomes

By the end of the course, students will be able to achieve the following objectives.

#	Outcome	Description
1	Model physical systems	Translate a physical system into a control-ready model with explicit assumptions and validity ranges, including actuator and sensor non-idealities
2	Design feedback controllers	Design baseline feedback controllers and select appropriate control architectures for real systems
3	Evaluate performance	Evaluate performance using standard time-domain metrics and interpret stability evidence in both time and frequency perspectives
4	Implement digitally	Implement controllers in discrete time and anticipate common sim-to-hardware failure modes
5	Handle nonlinearities	Model and reason about nonlinear effects and robustness
6	Work with multivariable systems	Represent and reason about multivariable systems in state space and understand the motivation for native multivariable controller designs
7	Communicate effectively	Communicate engineering decisions and experimental evidence in written and oral form

Project Team Members & Selection

Projects constitute a significant portion of the course grade and provide hands-on experience with control system design, implementation, and validation. Students will complete two projects throughout the semester, progressing from single-input single-output (SISO) systems to more complex multi-input multi-output (MIMO) systems with machine learning integration.

All projects must be completed in teams of 2–3 students. Single-person projects are not permitted due to insufficient time for individual presentations and the collaborative nature of real-world engineering practice.

Students may self-organize into teams during the first lecture. After the first lecture, any remaining unassigned students will be placed into teams by the instructor. No team changes are permitted until after Project 1 is complete. Requests for changes after Project 1 must be submitted in writing.

Project 1: SISO Control Implementation

Project 1 focuses on the design and implementation of a controller for a single-input single-output (SISO) system. This project establishes foundational skills in control implementation that will be extended in Project 2.

Systems may be continuous-time or discrete-time. Implementation must be on real hardware or a detailed simulation with realistic sensor and actuator models (including noise, quantization, delays, and saturation). Projects are open-ended—students are encouraged to propose their own systems.

The following systems are provided as a starting menu. Students may select from this list or propose an alternative system for approval.

System	Input	Output
Inverted pendulum / cart-pole	Cart force or motor voltage	Pendulum angle
Vehicle steering control	Steering angle	Lateral position or heading
DC motor position control	Motor voltage	Shaft position
DC motor speed control	Motor voltage	Shaft angular velocity
Ball and beam	Beam angle	Ball position
Thermal system regulation	Heater power	Temperature
<i>Custom approved system</i>	<i>TBD</i>	<i>TBD</i>

Project 2: MIMO with ML-Based Control

Project 2 advances to multi-input multi-output (MIMO) systems and introduces machine learning-based control methods. This project emphasizes the comparison between classical and modern control approaches.

Systems should have multiple inputs and outputs (preferably many I/O channels). Implementation must be discrete-time. Each project must include both an ML-based controller (e.g., reinforcement learning, neural network control, adaptive methods) and a classical controller for comparison (e.g., LQR, PID, state feedback). Projects are open-ended—students are encouraged to continue/expand their Project 1 system or propose a new system.

Project Resources

The following resources are available to all project teams.

System	Inputs	Outputs
Quadrotor attitude/position	Motor speeds (4)	Roll, pitch, yaw, position (6)
Multi-joint robotic arm	Joint torques/voltages	Joint angles, end-effector pose
Multi-zone HVAC system	Damper positions, fan speeds	Zone temperatures, humidity
Autonomous vehicle	Steering, throttle, braking	Position, velocity, heading
<i>Custom approved system</i>	<i>TBD</i>	<i>TBD</i>

Resource	Details
3D Printers	Available for fabricating custom mechanical parts
Computers	Workstations with full course software environment
Arduino Microcontrollers	Standard issue for all teams
Raspberry Pi	Available upon request with written justification of computational requirements
Sensor/Actuator Kits	Standardized kits provided to each team

Additional Options:

Personal Equipment	Students may purchase additional equipment at their own expense
Research Lab Equipment	Students may integrate equipment from their research laboratories with advisor approval

Project Demonstrations

All projects require demonstration of the implemented control system. A recorded video demonstration is required and must be uploaded with the project submission. Live demonstrations during presentations are permitted only if the equipment is small and portable; large or fragile setups should be demonstrated via video only.

Demonstrations must show multiple trials to establish repeatability of performance—a single successful run is insufficient. Demonstrations must also include failure case analysis: What conditions cause the controller to fail? What are the stability boundaries? How does the system degrade under disturbances or parameter variations?

Project Deliverables

Both projects require the following deliverables. All items are mandatory unless explicitly stated otherwise.

#	Deliverable	Description
1	Project Proposal Document	Submitted for instructor approval <i>before</i> work begins. Must include system description, objectives, methodology, and timeline.
2	Code Repository	All source code must be submitted. Code must be reproducible with clear instructions for execution.
3	Team Report	Comprehensive technical report covering theory, design, implementation, results, and analysis.
4	Individual Report	Each team member submits a personal reflection on their contributions and key learnings.
5	Presentation Slides	Submitted <i>before</i> presentation day for review.
6	Live Presentation	In-class presentation followed by Q&A session with instructor and peers.
7	Demo Video	Recorded demonstration with narration explaining the system operation and results.

Course Policies

AI Tools Policy

Artificial intelligence tools are allowed and highly encouraged in this course. Given the complexity of modern control system design and implementation, students who do not utilize AI assistance are likely to struggle significantly. AI may be used without restriction on all homework assignments, projects, and other coursework. Disclosure of AI use is not required and is discouraged. The only exception is examinations, where AI tools are not permitted.

Collaboration

Conceptual discussion with classmates is allowed and encouraged, but all submissions must be your own work. Project work must reflect equitable contribution from all team members. External resources must be cited.

Late Work

Late homework submissions are generally not accepted because one lowest score is dropped. Project deadlines are firm due to hardware scheduling and presentation logistics.

Accessibility

Students with accommodations should coordinate with the instructor early in the semester to ensure appropriate arrangements can be made.

Assignment Summary

Each homework (HW1–HW10) consists of two files: HW_ X_m (math) and HW_ X_c (code).

Assignment	Due Date	Topic Coverage
HW0	Sun 8/16 (midnight)	Mathematics Prerequisites
HW1	Mon 8/24	Modeling I (Lecture 1)
HW2	Mon 8/31	Modeling II: Actuators & Sensors (Lectures 2–3)
HW3	Tue 9/8	Basic Feedback Design (Lectures 4–5)
HW4	Mon 9/14	Stability Analysis (Lectures 6–7)
HW5	Mon 9/21	Architecture & State Space (Lectures 8–9)
HW6	Mon 10/5	Discrete & Nonlinear (Lectures 11–13)
HW7	Mon 10/12	State Estimation (Lecture 13)
HW8	Mon 10/26	MIMO Design (Lectures 16–18)
HW9	Mon 11/9	Learned Perception (Lectures 21–22)
HW10	Mon 11/23	Learned Control (Lectures 23–26)
Exam 1	Tue 9/22	Modeling, Feedback, Stability
Exam 2	Thu 10/29	MIMO, Robustness, Implementation
Project 1 Proposal	Thu 9/24	—
Project 1 Report	Thu 10/8	—
Project 2 Proposal	Tue 11/3	—
Project 2 Report	Thu 12/3	—

Course Schedule

The course meets Tuesdays and Thursdays throughout Fall 2026. The schedule below maps all 28 meetings to specific dates, accounting for university holidays and breaks.

Date	Lec	Type	Topic
Thu 8/20	1	Content	Modeling Physical Systems for Control
Tue 8/25	2	Content	Modeling Physical Actuators for Control
Thu 8/27	3	Content	Modeling Sensor Error for Control
Tue 9/1	4	Content	Designing Control Systems
Thu 9/3	5	Content	Measuring Controller Performance
Tue 9/8	6	Content	Stability Analysis in Time and Frequency Domain
Thu 9/10	7	Content	Control Architecture
Tue 9/15	8	Content	Introducing State Space Representation
Thu 9/17	9	Buffer	Integration Studio and Exam 1 Prep
Tue 9/22	10	Exam	Exam 1: Modeling, Feedback, Stability
Thu 9/24	11	Content	Implementing Control in Discrete Domain
Tue 9/29	12	Content	Nonlinear System Accounting and Control Robustness
Thu 10/1	13	Content	Estimating State in Nonlinear Systems
Tue 10/6	14	Presentation	Project 1 Presentations Day 1
Thu 10/8	15	Presentation	Project 1 Presentations Day 2
Tue 10/13	16	Content	Modeling MIMO Systems in State Space
Thu 10/15	—	<i>No Class</i>	<i>Fall Break</i>
Tue 10/20	17	Content	Designing Control Systems for MIMO
Thu 10/22	18	Content	Native MIMO Controllers: LQR, LQG, MPC
Tue 10/27	19	Buffer	Integration Studio and Exam 2 Prep
Thu 10/29	20	Exam	Exam 2: MIMO, Robustness, Implementation
Tue 11/3	21	Content	CNN and Transformer Encoders for State Estimation
Thu 11/5	22	Content	Transformer World Models for Dynamics
Tue 11/10	23	Content	Hybrid Model-Based and Learned Control
Thu 11/12	24	Content	Differentiable Predictive Control
Tue 11/17	25	Content	Diffusion Policies for Trajectory Tracking
Thu 11/19	26	Content	Generalization and Sim-to-Real
Tue 11/24	—	<i>No Class</i>	<i>Thanksgiving Break</i>
Thu 11/26	—	<i>No Class</i>	<i>Thanksgiving Break</i>
Tue 12/1	27	Presentation	Final Project Presentations Day 1
Thu 12/3	28	Presentation	Final Project Presentations Day 2